Numerical Solutions to the 2D Solar Forced Temperature Evolution of Fresh Water Lakes

E M A/E P 471 Final Report

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1 Introduction

In the wintertime, freshwater lakes in the all over the country freeze over, creating an ice layer above the lakes. The Midwest is no exception to this. This becomes a major area of interest since Lake Michigan, and more locally, Lake Mendota & Monona are home to many beloved ice activities. These include, but are not limited to, ice fishing, ice skating, and ice hockey. This means lake safety becomes a relevant issue. Weather during the winter can vary, but there is no doubt that sunlight will still hit these icy lakes, speeding up the melting process.

A consequence of the sunlight, is the process that occurs just beneath the ice layer itself. The rays from the sun can permeate through the ice, heating the water just beneath the ice layer. The density of freshwater has the property where its density will increasing with temperature until $\sim 4°\text{C}$, where it reaches a maximum of $\sim 1000\ \text{kg}\cdot\text{m}^{-3}$. After four degrees, the density decreases as the temperature increases. The water at the top of the lake is about zero degrees, which is just the temperature of frozen water. As the solar rays heat this bit of water below the ice, the density will increase until it hits four degrees. Since we are increasing the density of that parcel of water, it will want to sink towards the bottom of the lake. The less dense water beneath and surrounding it will have to replace this parcel by being forced upwards, creating a convective motion and an unstable water column. This process is known as radiatively driven convection (RDC).

In addition to recreation, we can use RDC to study biological processes within the lake itself. The RDC can cause a transport of very small life and nutrients, such as phytoplankton and algae. Limnologists can use RDC to predict the onset of algal blooms in lakes, which can cause the ice to melt even faster.

The study of RDC has been around for quite a while. Mathematically, we can model RDC and the temperature of these freshwater lakes using the Navier-Stokes (equation 1) and advective heat equations (equation 2).

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u} + \rho\mathbf{g} \tag{1}$$

$$\frac{\partial T}{\partial t} + \mathbf{u}\cdot\nabla T = \kappa\nabla^2 T + F(x,y,z) \tag{2}$$

where $F(x,y,z)$ is some forcing function. In the case of RDC, this will represent solar forcing. The 1987 paper by Matthews and Heaney [2] finds an analytical solution for the non-advective temperature profile $T(z,t)$ of as it varies with depth and time.

$$\begin{aligned} T(z,t) = {} & \underbrace{\left(\frac{C}{K\varepsilon_2^2} - A\right)\operatorname{erfc}\left(\frac{z}{2\sqrt{Kt}}\right)}_{\text{term 1}} \\ & \underbrace{- \frac{C}{2K\varepsilon_2^2}\exp\left(K\varepsilon_2^2 t\right)\left[\exp(-\varepsilon_2 z)\operatorname{erfc}\left(\frac{z}{\sqrt{Kt}} - \varepsilon_2\sqrt{Kt}\right) + \exp(\varepsilon_2 z)\operatorname{erfc}\left(\frac{z}{\sqrt{Kt}} + \varepsilon_2\sqrt{Kt}\right)\right]}_{\text{term 2}} \\ & \underbrace{+ \frac{C}{K\varepsilon_2^2}\exp(-\varepsilon_2 z)\left[\exp(\varepsilon_2^2 Kt) - 1\right]}_{\text{term 3}} + \underbrace{A + Bz}_{\text{term 4}} \end{aligned} \tag{3}$$

Each term in equation 3 has been labeled as its been split up in the `T_analytic` function in the main python file `main.py`. For the one-dimensional profiles without advection we will use the solar-forcing function defined in Smith et al (2025).

$$F(z) = \frac{Q_0}{\lambda}e^{z/\lambda} \tag{4}$$

where the radiative forcing is defined as

$$Q_0 = \frac{I_0}{\rho_{\text{MD}} c_p} \tag{5}$$

such that the light intensity I_0 is given by

$$I_0(p) = (1-\alpha) I_S \exp\left(-\left[k_w p + k_b(1-p)\right] L\right) \tag{6}$$

This tells us that the solar-forcing function is dependent on the depth (z), light attenuation (λ), max density of water ($\rho_{\text{MD}} \approx 999.97$ kg m^{-3}), heat capacitance of water ($c_p = 4210$ J K^{-1}C^{-1}), proportion of opaque "white" ice to translucent "black" ice (p), attenuation coefficients of white (k_w) and black (k_b) ice, surface albedo (α), and length of the ice layer (L). These values can all be determined experimentally. We will adopt the values used in the supporting information from Smith et al (1987) [3]. The Smith and Matthews and Heaney papers use different notation, here we will adopt Smith's. Table 1 gives the relation between the different notation.

Variable	**Matthews & Heaney**	**Report**	**Code**
Diffusivity	K	κ	`K`
Diffusive length	$\sqrt{Kt}$	L_{diff}	`L_diff`
Water clarity	ε_2	$1/\lambda$	`e2`
Free surface temperature	A	T_∞	`A`
Scaled forcing	C	Q_0/λ	`C`
Initial temperature gradient	B	T_H/H	`B`

Table 1: Nomenclature for variables used to define analytical solution in `main.py`.

The first part of this report is to use numerical methods to confirm the analytical solution from Matthews and Heaney by approximating the advective heat equation for one-dimensional and two-dimensional cases in the absence of advection, i.e. $\mathbf{u} = 0$. This part will use the one-dimensional solar-forcing function defined by Smith. The second part is observe the temperature evolution in the presence of advection through contour plots. In order to achieve this, we will need a forcing function that also include a horizontal (x) component as well. Allum (2023)[1] uses the following direction for the forcing function which is based on the shadow created by snow pattern on the ice layer.

$$F(x,z) = \exp\left(-\frac{L_z - z}{\lambda}\right)\left[1 - A_l \exp\left(-\frac{(x - x_s)^{8w_d}}{w_d^{8w_d}}\right)\right] \tag{7}$$

The first exponential term is mathematically equivalent to what we originally defined for the solar-forcing function. Also, there are three new parameters: snow surface albedo (A_l), shadow position (x_s), and shadow width (w_d). So, our forcing function will take the form,

$$F(x,z) = \exp\left(\frac{z}{\lambda}\right)\left[1 - A_l \exp\left(-\frac{(x - x_s)^{8w_d}}{w_d^{8w_d}}\right)\right] \tag{8}$$

This should give us the shape of a "super" Gaussian, which we will observe in the results and mesh convergence sections.

2 Finite Difference Derivation

As mentioned in the introduction, all of the following derivations will be based on the solar-forced heat equation.

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T = \kappa \nabla^2 T + F(x,z)$$

Where for the non-advective one-dimensional and two-dimensional cases we will use equation 4 and for the advective (contour) cases we will use equation 8. We will also use the following finite difference method (FDM) approximations for center in space (CS) and forward in time (FT).

$$\frac{\partial T}{\partial t} = \frac{T_{i,j}^{k+1} - T_{i,j}^{k}}{\Delta t} \tag{9}$$

$$\frac{\partial T}{\partial x} = \frac{T_{i+1,j}^{k} - T_{i-1,j}^{k}}{2\Delta x}, \quad \frac{\partial T}{\partial z} = \frac{T_{i,j+1}^{k} - T_{i,j-1}^{k}}{2\Delta z} \tag{10}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{T^k_{i+1,j} - 2T^k_{i,j} + T^k_{i-1,j}}{\Delta x^2}, \quad \frac{\partial^2 T}{\partial z^2} = \frac{T^k_{i,j+1} - 2T^k_{i,j} + T^k_{i,j-1}}{\Delta z^2} \tag{11}$$

Our goal is to start at the most basic form: one-dimensional and no advection and work our way to a more realistic scenario: two-dimensional with uniform velocity.

2.1 One-dimensional

First we will consider the one-dimensional case where there is only a temperature gradient in the z-direction and no advective velocity ($\mathbf{u} = \mathbf{0}$). This implies that the heat equation reduces to,

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial z^2} + \frac{Q_0}{\lambda} e^{\frac{z}{\lambda}}$$

Applying FDM we get,

$$\frac{T^{k+1}_{i,j} - T^k_{i,j}}{\Delta t} = \kappa \frac{T^k_{i,j+1} - 2T^k_{i,j} + T^k_{i,j-1}}{\Delta z^2} + \frac{Q_0}{\lambda} e^{\frac{z_j}{\lambda}}$$
$$T^{k+1}_{i,j} - T^k_{i,j} = \frac{\kappa \Delta t}{\Delta z^2} \left[T^k_{i,j+1} - 2T^k_{i,j} + T^k_{i,j-1}\right] + \frac{Q_0}{\lambda} e^{\frac{z_j}{\lambda}} \Delta t$$

such that we can define the diffusion number $r = \frac{\kappa \Delta t}{\Delta z^2}$.

$$T^{k+1}_{i,j} = rT^k_{i,j+1} + (1-2r)T^k_{i,j} + rT^k_{i,j-1} + \frac{Q_0}{\lambda} e^{\frac{z_j}{\lambda}} \Delta t$$

Additionally, for notion purposes, we will define the forcing-number as $Q'_j = \frac{Q_0}{\lambda} e^{\frac{z_j}{\lambda}} \Delta t$, thus arriving at

$$T^{k+1}_{i,j} = rT^k_{i,j+1} + (1-2r)T^k_{i,j} + rT^k_{i,j-1} + Q'_j \tag{12}$$

We do not need to consider the advective form of the one-dimensional solar-forcing heat equation since throughout this report, we will assume velocity only has an x-component, and we are observing the diffusion in the z-direction.

$$\mathbf{u} \cdot \nabla T = U \hat{\mathbf{x}} \cdot \hat{\mathbf{z}} \frac{\partial T}{\partial z} \rightarrow \hat{\mathbf{x}} \cdot \hat{\mathbf{z}} = 0$$

2.2 Two-dimensional without advection

Recall the solar-forced heat equation,

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T = \kappa \nabla^2 T + \frac{Q_0}{\lambda} e^{\frac{z}{\lambda}}$$
$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \left(\hat{\mathbf{x}} \frac{\partial T}{\partial x} + \hat{\mathbf{z}} \frac{\partial T}{\partial z} \right) = \kappa \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2} \right) + \frac{Q_0}{\lambda} e^{\frac{z}{\lambda}}$$

Since we are considering the non-advective case ($\mathbf{u}$=0), the PDE becomes parabolic and significantly easier to solve.

$$\frac{\partial T}{\partial t} = \kappa \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2} \right) + \frac{Q_0}{\lambda} e^{\frac{z}{\lambda}}$$

And now applying FDM,

$$\frac{T^{k+1}_{i,j} - T^k_{i,j}}{\Delta t} = \kappa \left(\frac{T^k_{i+1,j} - 2T^k_{i,j} + T^k_{i-1,j}}{\Delta x^2} + \frac{T^k_{i,j+1} - 2T^k_{i,j} + T^k_{i,j-1}}{\Delta z^2} \right) + \frac{Q_0}{\lambda} e^{\frac{z_j}{\lambda}}$$
$$\frac{T^{k+1}_{i,j} - T^k_{i,j}}{\Delta t} = \kappa \frac{T^k_{i+1,j} - 2T^k_{i,j} + T^k_{i-1,j}}{\Delta x^2} + \kappa \frac{T^k_{i,j+1} - 2T^k_{i,j} + T^k_{i,j-1}}{\Delta z^2} + \frac{Q_0}{\lambda} e^{\frac{z_j}{\lambda}}$$
$$T^{k+1}_{i,j} - T^k_{i,j} = \frac{\kappa \Delta t}{\Delta x^2} \left(T^k_{i+1,j} - 2T^k_{i,j} + T^k_{i-1,j}\right) + \frac{\kappa \Delta t}{\Delta z^2} \left(T^k_{i,j+1} - 2T^k_{i,j} + T^k_{i,j-1}\right) + \frac{Q_0}{\lambda} e^{\frac{z_j}{\lambda}} \Delta t$$

We define the diffusive numbers $r_x = \frac{\kappa \Delta t}{\Delta x^2}, r_z = \frac{\kappa \Delta t}{\Delta z^2}$.

$$T_{i,j}^{k+1} - T_{i,j}^k = r_x \left(T_{i+1,j}^k - 2T_{i,j}^k + T_{i-1,j}^k\right) + r_z \left(T_{i,j+1}^k - 2T_{i,j}^k + T_{i,j-1}^k\right) + Q_j'$$

$$T_{i,j}^{k+1} = (1 - 2(r_x + r_z))T_{i,j}^k + r_x T_{i+1,j}^k + r_x T_{i-1,j}^k + r_z T_{i,j+1}^k + r_z T_{i,j-1}^k + Q_j' \tag{13}$$

This will also mark the transition of the shift from depth-temperature profiles to contour plots show the temperature distribution due to solar-forcing and advection. That means we will now introduce the Allum forcing function to our FDM scheme. The only change will be with the Q_j' term. Instead, we can add the Gaussian term,

$$F(x,z) = \frac{Q_0}{\lambda} \exp\left(\frac{z}{\lambda}\right) \left[1 - A_l \exp\left(-\frac{(x - x_s)^{8w_d}}{w_d^{8w_d}}\right)\right] \tag{14}$$

we include the Q_0/λ term again because the Allum paper states that this is the magnitude of the forcing function, which means the exponential purely gives the direction/shape. We can discretize to get,

$$F_{i,j} = \frac{Q_0}{\lambda} \exp\left(\frac{z_j}{\lambda}\right) \left[1 - A_l \exp\left(-\frac{(x_i - x_s)^{8w_d}}{w_d^{8w_d}}\right)\right] \Longrightarrow F_{i,j}' = F_{i,j} \Delta t \tag{15}$$

For our contours, we will use,

$$T_{i,j}^{k+1} = (1 - 2(r_x + r_z))T_{i,j}^k + r_x T_{i+1,j}^k + r_x T_{i-1,j}^k + r_z T_{i,j+1}^k + r_z T_{i,j-1}^k + F_{i,j}' \tag{16}$$

2.3 Upwind scheme

For the next two cases we will consider constant and uniform advection. Basic finite difference methods become more unstable in the presence of advection, so we have to choose a similar numerical method. The upwind scheme is another form of the FDM we have already used, except it combines forward and backward differences to use information about where the flow is coming from. This is a popular method used in computational fluid dynamics, especially in advection cases. The only downside to the upwind scheme is that is sacrifices some of the diffusive accuracy given by the rest of the differential equation. The scheme is defined as follows,

$$\frac{\partial T}{\partial x} \approx \begin{cases} \frac{T_{i,j} - T_{i-1,j}}{\Delta x}, & U > 0 \\ \frac{T_{i+1,j} - T_{i,j}}{\Delta x}, & U < 0 \end{cases} \tag{17}$$

Where U is the fluid's velocity. For simplicity of notation in the derivations, we will write each as

$$U > 0: \quad T_x^+ = \frac{T_{i,j} - T_{i-1,j}}{\Delta x}$$
$$U < 0: \quad T_x^- = \frac{T_{i+1,j} - T_{i,j}}{\Delta x}$$

We do not need to apply this to the two-dimensional non-advection case since there is no advective term to apply this scheme to.

2.4 Two-dimensional with constant advection

We will now introduce a constant one-dimensional advective velocity $\mathbf{u} = U\hat{\mathbf{x}}$.

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T = \kappa \nabla^2 T + F(x,y)$$
$$\frac{\partial T}{\partial t} + (U\hat{\mathbf{x}}) \cdot \left(\hat{\mathbf{x}} \frac{\partial T}{\partial x} + \hat{\mathbf{z}} \frac{\partial T}{\partial z}\right) = \kappa \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2}\right) + F(x,z)$$
$$\frac{\partial T}{\partial t} + U \frac{\partial T}{\partial x} = \kappa \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2}\right) + F(x,z)$$

Which, using the upwind scheme, is discretized to,

$$\frac{T_{i,j}^{k+1}-T_{i,j}^{k}}{\Delta t}+U\left(T_x^+ + T_x^-\right)=\kappa\left(\frac{T_{i+1,j}^{k}-2T_{i,j}^{k}+T_{i-1,j}^{k}}{\Delta x^2}+\frac{T_{i,j+1}^{k}-2T_{i,j}^{k}+T_{i,j-1}^{k}}{\Delta z^2}\right)+F_{i,j}$$

$$\frac{T_{i,j}^{k+1}-T_{i,j}^{k}}{\Delta t}=-U\left(T_x^+ + T_x^-\right)+\kappa\left(\frac{T_{i+1,j}^{k}-2T_{i,j}^{k}+T_{i-1,j}^{k}}{\Delta x^2}+\frac{T_{i,j+1}^{k}-2T_{i,j}^{k}+T_{i,j-1}^{k}}{\Delta z^2}\right)+F_{i,j}$$

For $U > 0$:

$$T_{i,j}^{k+1}=(1-r_U-2(r_x+r_z))T_{i,j}^{k}+(r_U+r_x)T_{i-1,j}^{k}+r_xT_{i+1,j}^{k}+r_zT_{i,j+1}^{k}+r_zT_{i,j-1}^{k}+F'_{i,j} \tag{18}$$

For $U < 0$:

$$T_{i,j}^{k+1}=(1+r_U-2(r_x+r_z))T_{i,j}^{k}+r_xT_{i-1,j}^{k}+(r_x-r_U)T_{i+1,j}^{k}+r_zT_{i,j+1}^{k}+r_zT_{i,j-1}^{k}+F'_{i,j} \tag{19}$$

such that we define the advective stability parameter $r_U = \frac{U\Delta t}{\Delta x}$.

2.5 Two-dimensional with uniform advection

Finally we will consider a uniform velocity given by $\mathbf{u} = \left(U\frac{H-z}{H}, 0\right)$. This will have a very similar discretized equation as 2.4.

$$\frac{\partial T}{\partial t}+\mathbf{u}\cdot\nabla T=\kappa\nabla^2 T+F(x,z)$$

$$\frac{\partial T}{\partial t}+\left(\frac{U}{h}z\,\hat{\mathbf{x}}\right)\cdot\left(\hat{\mathbf{x}}\frac{\partial T}{\partial x}+\hat{\mathbf{z}}\frac{\partial T}{\partial z}\right)=\kappa\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial z^2}\right)+F(x,z)$$

$$\frac{\partial T}{\partial t}+\frac{U}{h}z\frac{\partial T}{\partial x}=\kappa\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial z^2}\right)+F(x,z)$$

$$\frac{T_{i,j}^{k+1}-T_{i,j}^{k}}{\Delta t}=-\frac{U}{h}(T_x^+ + T_x^-)+\kappa\left(\frac{T_{i+1,j}^{k}-2T_{i,j}^{k}+T_{i-1,j}^{k}}{\Delta x^2}+\frac{T_{i,j+1}^{k}-2T_{i,j}^{k}+T_{i,j-1}^{k}}{\Delta z^2}\right)+F_{i,j}$$

For $U > 0$:

$$T_{i,j}^{k+1}=(1-r'_U-2(r_x+r_z))T_{i,j}^{k}+(r'_U+r_x)T_{i-1,j}^{k}+r_xT_{i+1,j}^{k}+r_zT_{i,j+1}^{k}+r_zT_{i,j-1}^{k}+F'_{i,j} \tag{20}$$

For $U < 0$:

$$T_{i,j}^{k+1}=(1+r'_U-2(r_x+r_z))T_{i,j}^{k}+r_xT_{i-1,j}^{k}+(r_x-r'_U)T_{i+1,j}^{k}+r_zT_{i,j+1}^{k}+r_zT_{i,j-1}^{k}+F'_{i,j} \tag{21}$$

where we define the reduced advective stability parameter $r'_U = \frac{(U/h)\Delta t}{2\Delta x} = r_U/h$.

2.6 Stability metrics

For each method, there is an associated stability metric for which the numerical solution becomes unstable and oscillitory. They can be described as sums of the Courant numbers and stability parameters. Since we use a central difference scheme, they must be less than one half. However, for the upwind scheme and advective stability parameter r_U, we must divide by two to account for the forward/backward in space, i.e. only interacting with one node.

Method	Parameters	Metric
1D–no advection	r	$r \leq 1/2$
2D–no advection	r_x, r_z	$r_x + r_z \leq 1/2$
2D–constant advection	r_U, r_x, r_z	$\lvert r_U\rvert/2 + r_x + r_z \leq 1/2$
2D–uniform advection	r'_U, r_x, r_z	$\lvert r'_U\rvert/2 + r_x + r_z \leq 1/2$

Table 2: Conditions for stability for each defined method.

3 Application of Boundary Conditions

Throughout this project, we will consider the following boundary conditions: the initial temperature at the top of the frozen lake is zero degrees Celsius, which indicates the water is frozen, hence an ice layer, and the bottom of the lake (sediments) is at four degrees Celsius, which is the temperature of maximum density for water. Mathematically the Dirichlet boundary conditions can be expressed as,

$$T(t, x, z = 0) = 0°\text{C}, \quad T(t, x, z = H) = 4°\text{C}$$

For the x-direction, we will assume periodic boundary conditions such that the heat flux entering the left side of the lake is the same as the heat flux leaving the right side of the lake.

$$T(t, x = x_{\text{left}}, z) = T(t, x = x_{\text{right}}, z)$$
$$\frac{\partial T}{\partial x}(t, x = x_{\text{left}}, z) = \frac{\partial T}{\partial x}(t, x = x_{\text{right}}, z)$$

We also have an initial condition which describes the initial temperature of the lake at $t = 0$.

$$T(t = 0, x, z) = T_0(x, z) = T_{\text{free surface}} + \frac{T_{MD}}{H} z$$

3.1 One-dimensional

Recall our discretized form of the one-dimensional solar-forced equation with no advection.

$$T_{i,j}^{k+1} = rT_{i,j+1}^{k} + (1 - 2r)T_{i,j}^{k} + rT_{i,j-1}^{k} + Q'_j$$

Since there is no x-component in this reduced form, we only need to consider the $T(t, z = 0) = 0°\text{C}$ and $T(t, z = H) = 4°\text{C}$ boundary conditions. These boundary conditions tell us that for $j = 0$, $T_{i,0}^{k} = 4$ and $j = N$, $T_{i,N}^{k} = 0$ assuming we have $N + 1$ nodes.

3.2 Two-dimensional

Applying the boundary conditions to our three two-dimensional equations will be very similar to the one-dimensional case. From the periodic conditions we will have,

$$T_{0,j}^{k} = T_{N_x,j}^{k}$$

where N_x is one less than the total number of nodes for the x-direction. Then, from the fixed surface and bottom temperature,

$$T_{i,0}^{k} = 4, \quad T_{i,N_z}^{k} = 0$$

where N_Z is one less than the total number of nodes for the z-direction. These boundary conditions will be the same for all three two-dimensional conditions, regardless of the advection. We defined that the advection comes from the lake internally not the boundaries. This also means that when we observe the contours, the temperature leaving the lake should be the same as the temperature entering the lake.

4 Results

Using the methods described throughout the report, the discretized equations were implemented into a Python program using the Visual Studio Code (VSCode) IDE. The methods can be found in the file `main.py` and were accomplished with the NumPy, MatPlotLib, and SciPy libraries.

Through the program, the following parameters were kept constant throughout each trial.

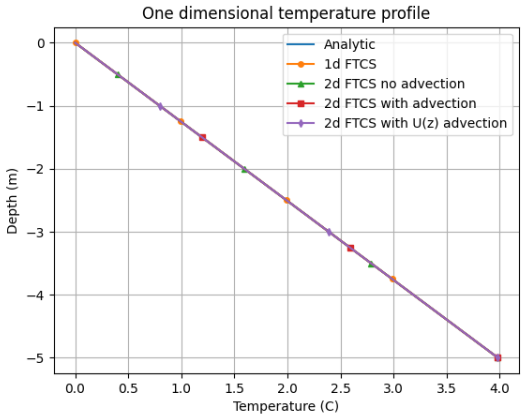
Variable	Symbol	Value	Code
Solar radiation	I_s	500 [Wm^{-2}]	`IS`
White ice attention	k_w	6.9 [m^{-1}]	`kw`
Black ice attention	k_b	2.2 [m^{-1}]	`kb`
Diffusivity of water	κ	1×10^{-7} [$\mathrm{m}^2\ \mathrm{s}^{-1}$]	`kappa`
Heat capacitance of water	c_p	4210 [$\mathrm{J\ kg}^{-1}\ {}^\circ\mathrm{C}^{-1}$]	`cP`
Max density of water	ρ_{MD}	999.97495 [$\mathrm{kg\ m}^{-3}$]	`rho_MD`
Temperature of max density	T_{MD}	3.983035 [°C]	`T_MD`
Max lake depth	H	5 [m]	`H`
Width of ice layer	L	0.5 [m]	`L`
Penetration depth	λ	0.1 [m^{-1}]	`lmbda`
Width of shadow	w_d	1 [m]	`wd`
Snow attenuation	A_l	1 [–]	`Al`

Table 3: Parameters used from Smith and Allum papers.

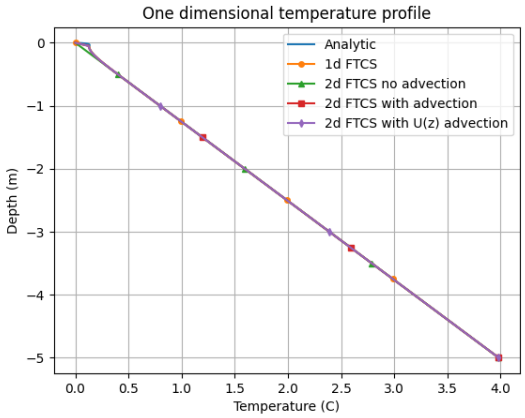
In addition, we will set the free surface temperature A to be zero degrees Celsius, which is the temperature of ice. We will also assume $p = 0.5 \implies \alpha = 0.3$ and $U = 0.01$ [$\mathrm{m\ s}^{-1}$]. We set the downward direction to be $+z$ over the domains $z \in [-H, 0]$ and $x \in [-H, 0]$.

4.1 Temperature-depth profiles

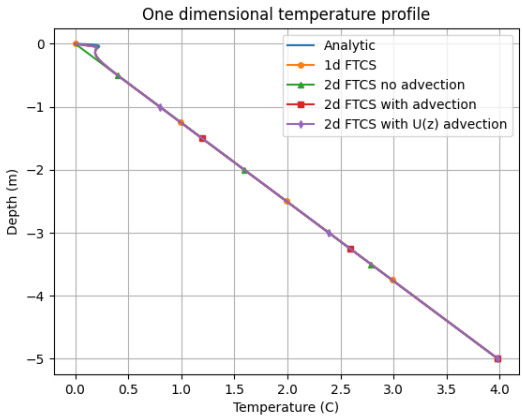
The first goal was to solve the FDM equations for all four cases and observe how accurately they match up to our analytical solution from Matthews and Heaney (1987). We will use 101 nodes and a time step of $dt = 1$ [s] and see how the curves evolve over $t \in \{0, 0.5, 1, 2, 4, 8\}$ [hr].



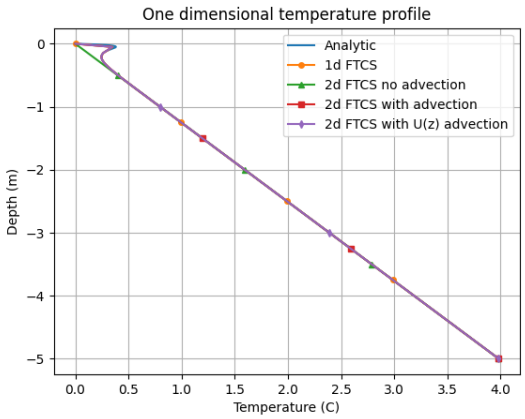
(a) $t = 0$ hr



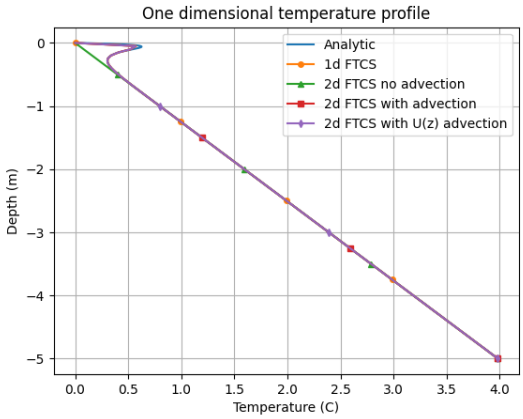
(b) $t = 0.5$ hr



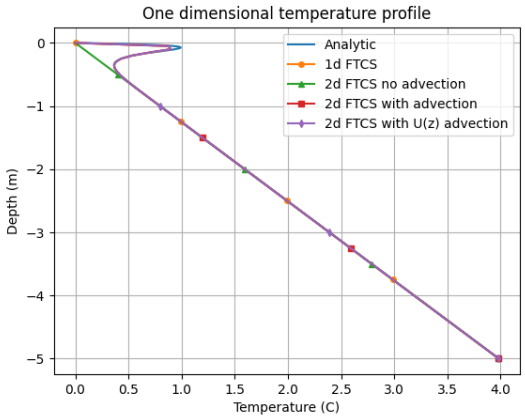
(c) $t = 1$ hr



(d) $t = 2$ hr



(e) $t = 4$ hr



(f) $t = 8$ hr

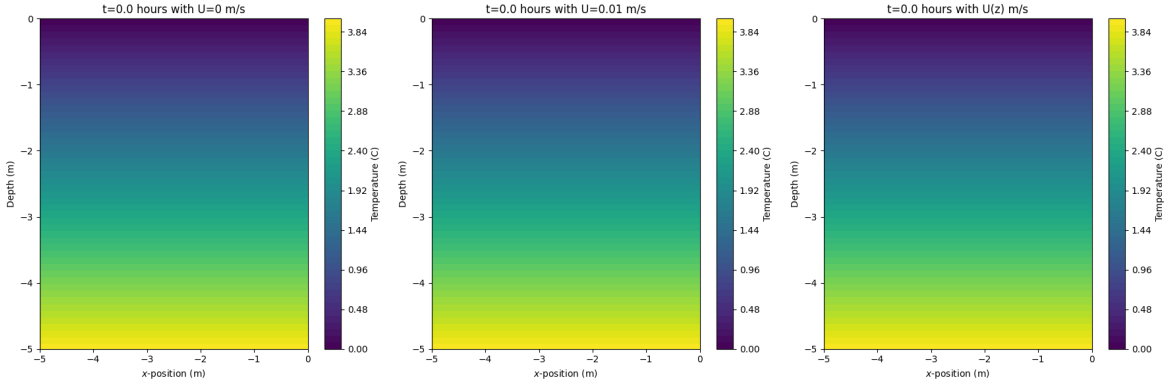
Figure 1: Evolution of the one-dimensional temperature profile due to solar forcing over time.

As we can tell from figure 1, the numerical solutions line up near perfectly with the analytical solution. Due to the difference scheme, while keeping stability in mind, the time step must be relatively big, or else the program would take a long time to solve. What we should be seeing on these graphs is an overhang

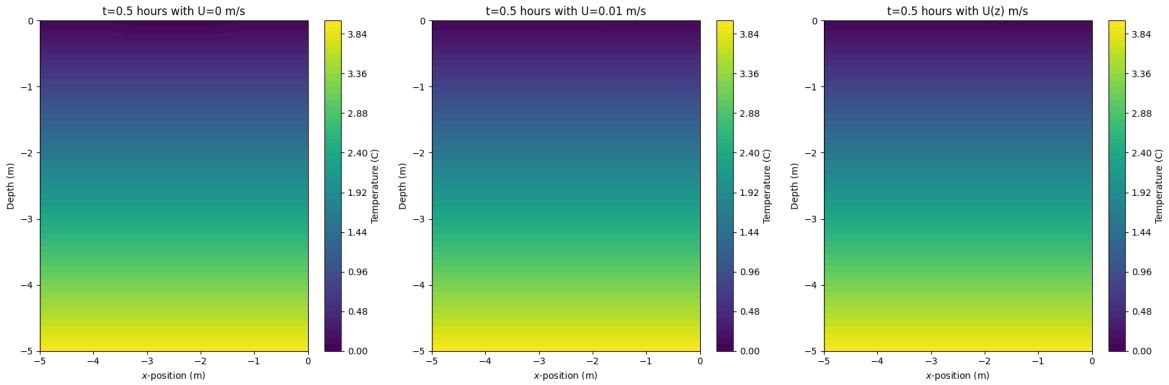
starting to form near the top. That is the RDC occurring in the thermocline layer of the lake (region of temperature change). These graphs also match up with the figures from Smith (2025) and Matthews and Heaney (1987).

4.2 Advection profiles

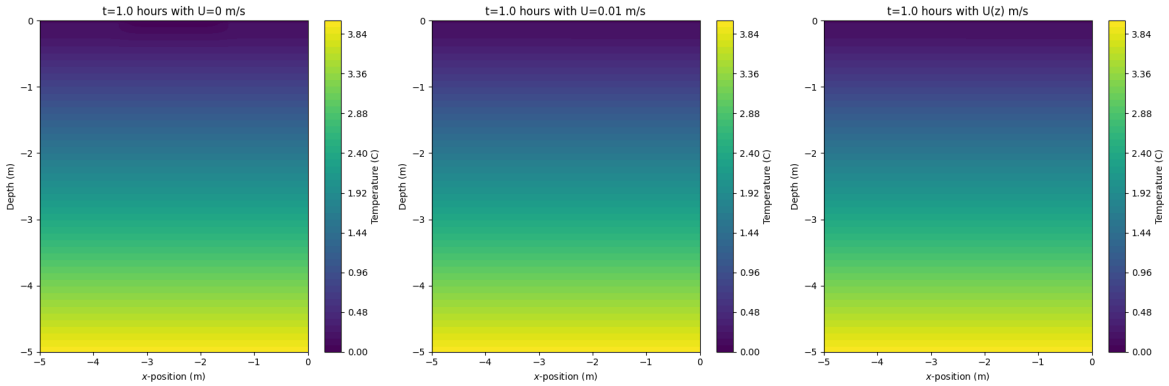
The next part of the project was to observe the overall 2D temperature distributions in the lake to demonstrate the effect of advection on this process.



(a) $t = 0$ hr

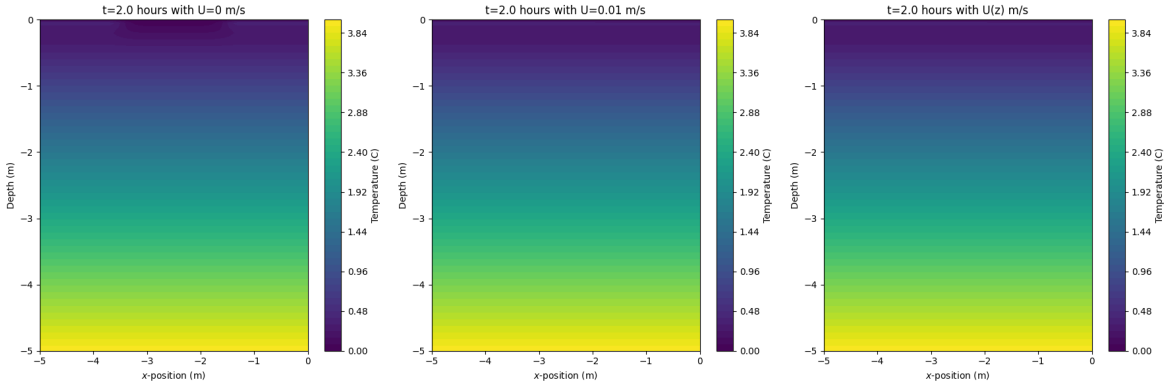


(b) $t = 0.5$ hr

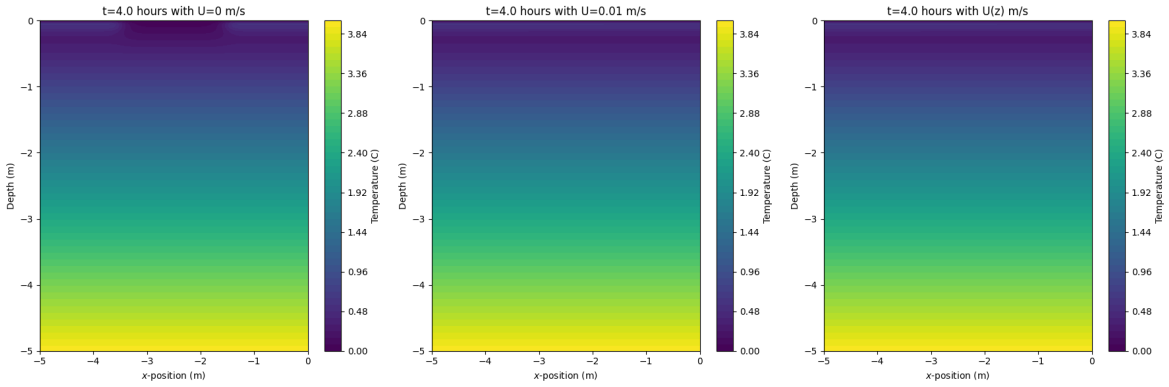


(c) $t = 1$ hr

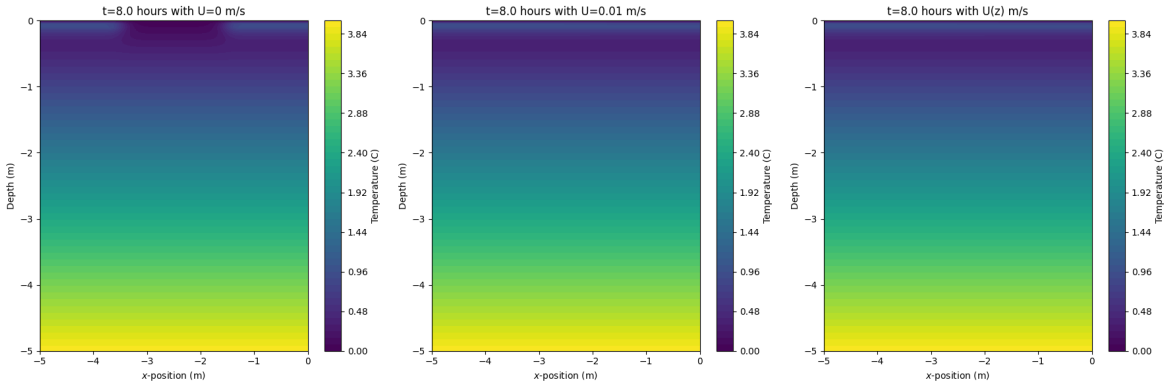
Figure 2: Evolution of the two-dimensional temperature contours due to solar forcing for early times.



(a) $t = 2$ hr



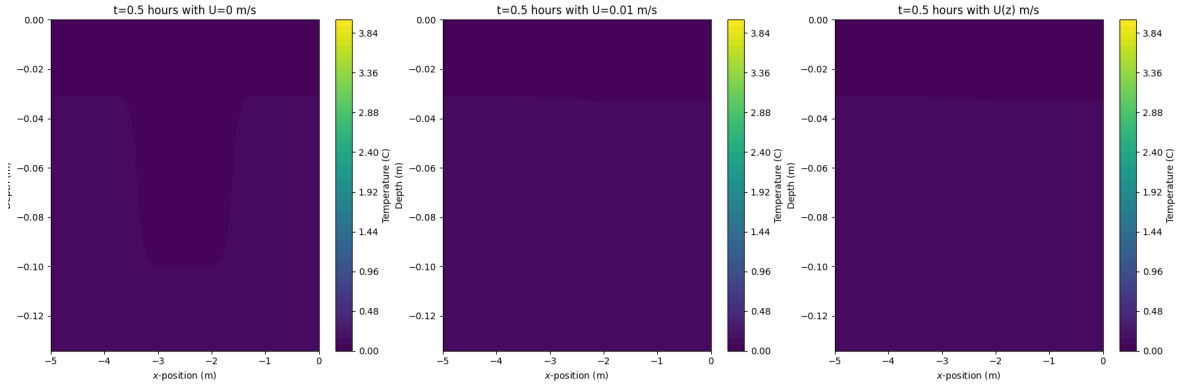
(b) $t = 4$ hr



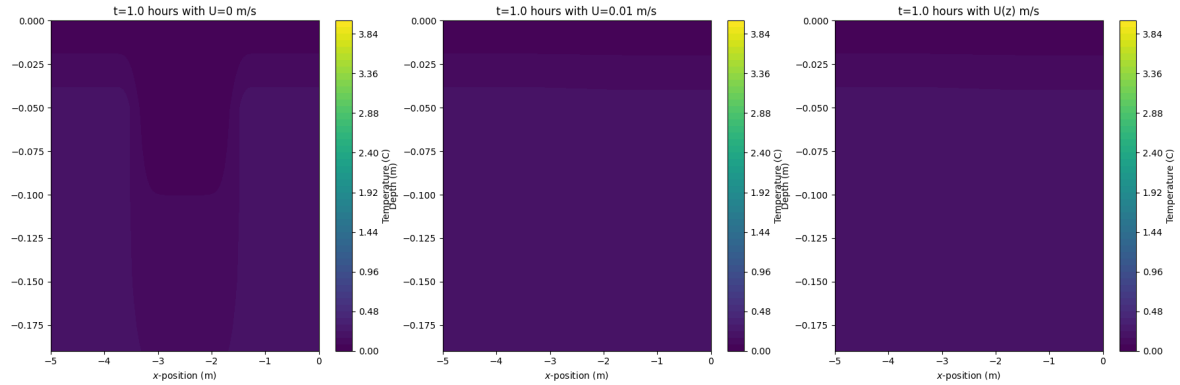
(c) $t = 8$ hr

Figure 3: Evolution of the two-dimensional temperature contours due to solar forcing for later times.

In these contours, it is very difficult to see the actual diffusion and advection that occurs since it happens at the top level within a few tenths of a meter, known as the diffusive length. We can set the limits of our contours to this diffusive scale and properly see the temperature perturbations.

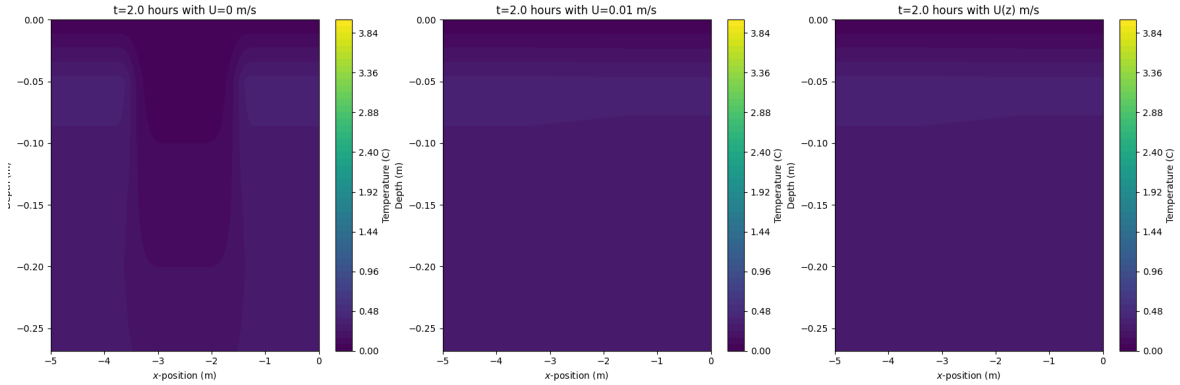


(a) $t = 0.5$ hr

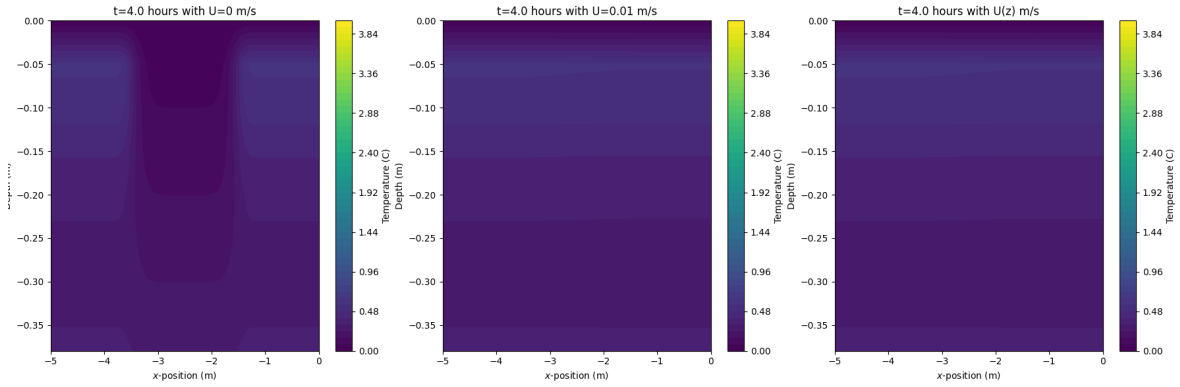


(b) $t = 1$ hr

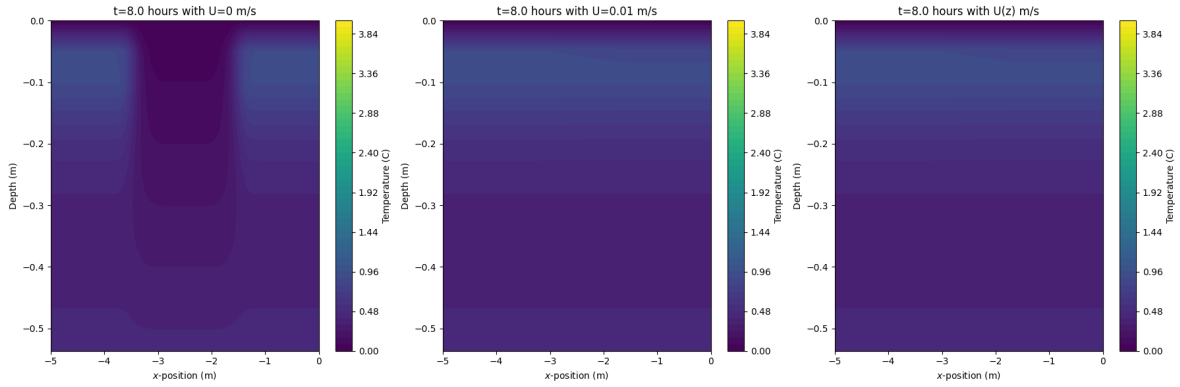
Figure 4: Diffusive region of the lake at earlier times.



(a) $t = 2$ hr



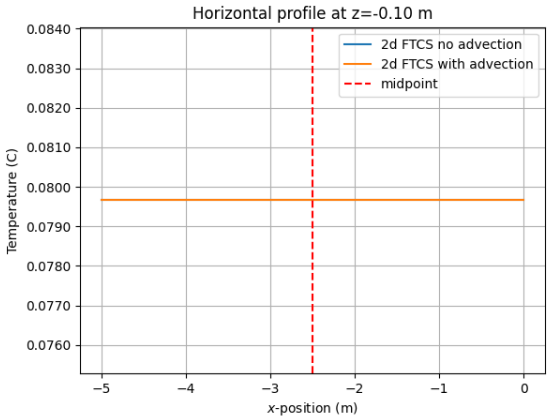
(b) $t = 4$ hr



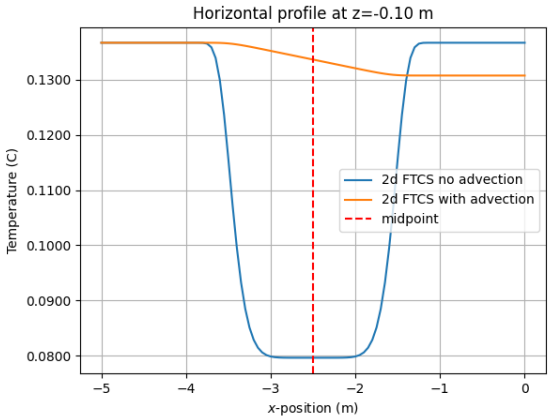
(c) $t = 8$ hr

Figure 5: Diffusive region of the lake at earlier times.

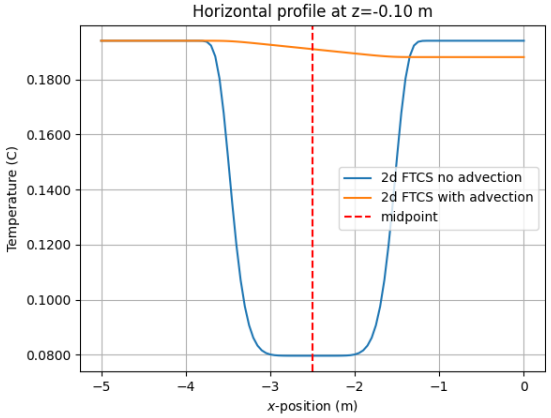
Now this temperature change is more defined. Without advection (left contour), we just see the diffusion from the sun. This forces the temperature downward into the lake in the "super" Gaussian shape mentioned earlier. As we add advection, the flow moves this temperature along in the x-direction which is then carries through by the periodic boundary conditions, hence why it seem like there is little to no temperature change. To get a better look at the effect of the advection, we can observe the temperature in the x-direction at $z = -0.1$ m, which is about the depth of the max light penetration depth.



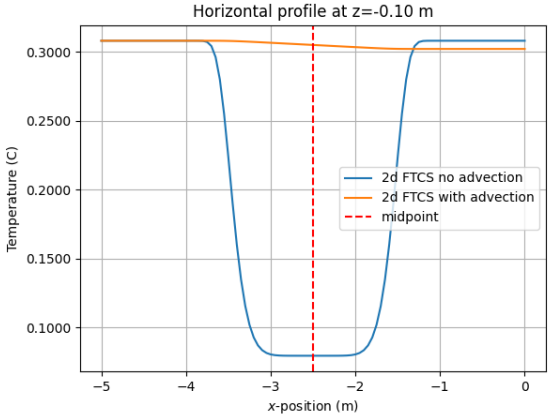
(a) $t = 0$ hr



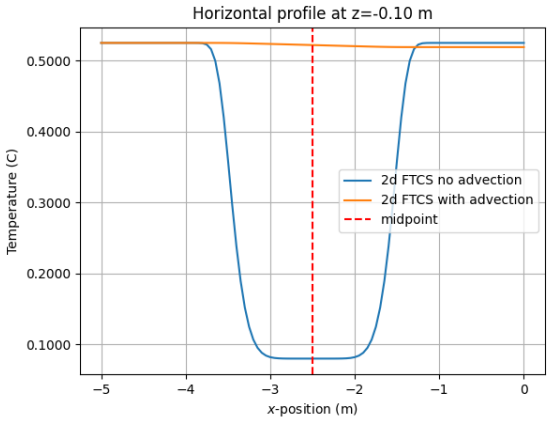
(b) $t = 0.5$ hr



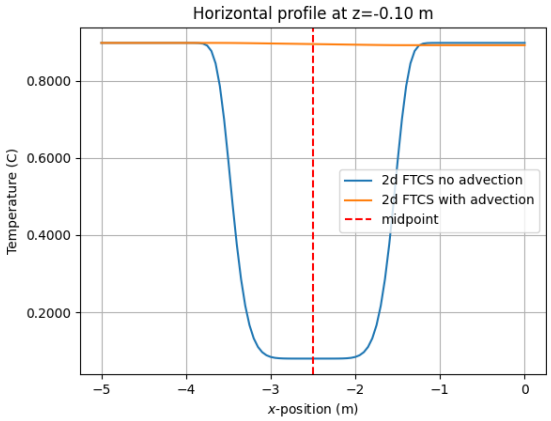
(c) $t = 1$ hr



(d) $t = 2$ hr



(e) $t = 4$ hr



(f) $t = 8$ hr

Figure 6: Effect of uniform advection on the temperature distribution over the lake.

These graphs show the non-advective case against the uniform advective case. Here, we can see that the advection does have a role, just very minor. The velocity take a Couette-esque flow shape such that it should increase linearly downwards, causing the temperature to slowly decrease, which the graphs show. The

non-advection case confirms what we see in the contour. It is important to recognize that the advection curve is not decreasing, but the non-advection curve becomes steeper, increasing the range of the plot, making it appear to get smaller.

5 Convergence Analysis

Although the produced graphs seemed accurate, both mathematically and physically, we still need to run an analysis on the actual accuracy and efficiency.

5.1 Time-Step Analysis

We can conduct an error convergence analysis for the time-step over $dt \in \{0.25, 0.5, 1, 2, 4\}$ seconds. Plotting the error we see a flat line.

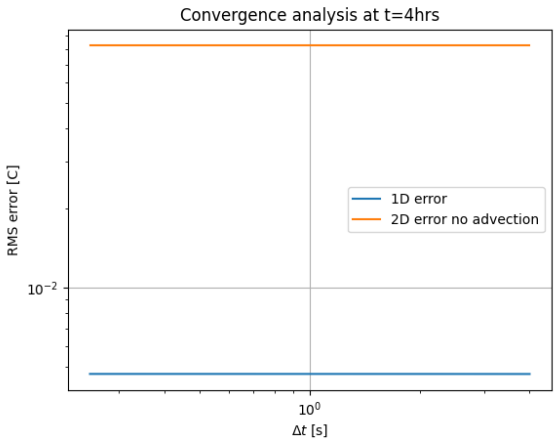
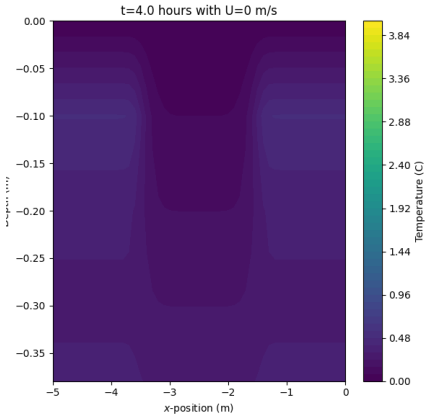


Figure 7: Time step error convergence analysis.

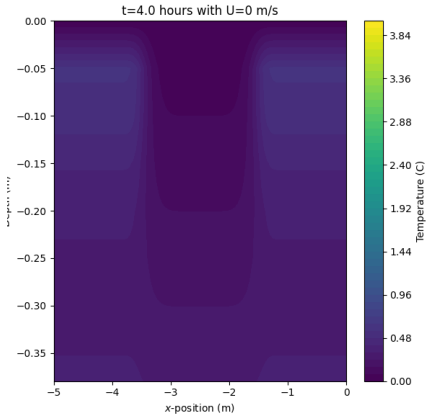
We can only plot the 1D error and 2D error with no advection because those are the only two curves that have the same exact forcing function and physics. The other two are influenced by advection and a different-shaped solar forcing function, which means it will produce different results. However, the curve appears flat for two reasons: boundary conditions and the effect of time-step. The paper which these models are based from may differ in boundary conditions and originate from different places which don't exactly line-up with what we have. Additionally, these plots are more dominated by the spatial resolution i.e. by Δz, Δx, and N. So it would be helpful to do a mesh/spatial analysis.

5.2 Mesh Analysis

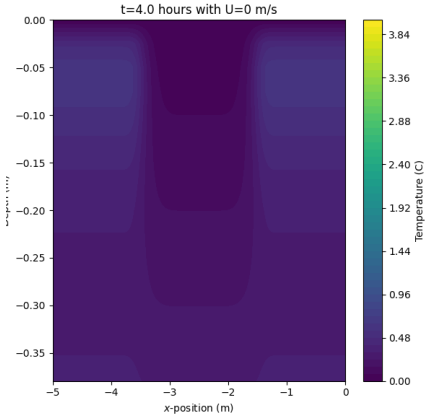
In computational fluid dynamics, the quality of your mesh is very important. Although we have produced a very basic model, we can vary the number of nodes we have to see how it affects our contour/simulation. We will consider the non-advective curve at $t = 4$ hours with a time step of $dt = 1$ second varying over $N \in \{51, 101, 201, 401, 801\}$.



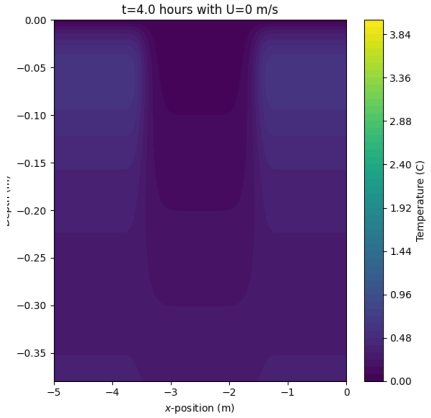
(a) 51 nodes



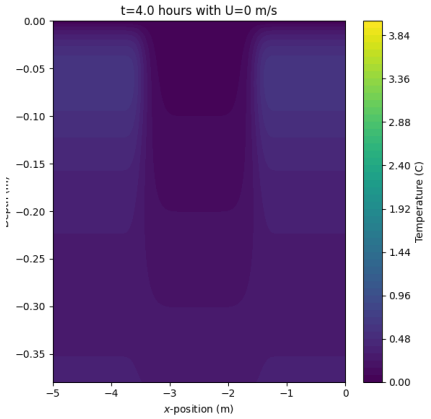
(b) 101 nodes



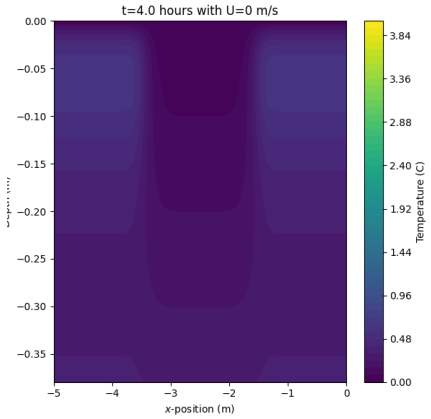
(c) 201 nodes



(d) 401 nodes



(e) 601 nodes



(f) 801 nodes

Figure 8: Mesh analysis with varying spatial resolutions for the non-advective contour at $t = 4$ hr.

From the contours, it seems like not much has changed, but if we look closely, we can see that the temperature becomes more defined. If we were to change the dz as well, but enough so the FDM is still stable this process would take a significantly longer time to process, but would see a clear difference.

6 Future Work

Throughout this report, we implemented finite difference schemes to solve the advective solar forced heat equation. More specifically, forward in time, center in space and the upwind scheme methods. Considering time and resources, the next steps would be to do a higher-quality spatial resolution analysis. Based on my computer it took hours to generate the plots based on the way the methods were programmed.

Additionally, one could approximate the Navier-Stokes equation instead. This could provide a more "exact" solution since it more properly accounts for the advection of the lake.

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u} + \rho\mathbf{g}$$

This would be significantly difficult since the water becomes highly turbulent, which means that the equation is dominated by the non-linear term $\mathbf{u}\cdot\nabla\mathbf{u}$. Overall, in this case, the finite difference methods do a good job in predicting the effect of the solar-forced heat equation in freshwater lakes.

References

[1] Donovan Allum. Simulations of Radiatively Driven Convection and Spatially Heterogeneous Solar Radiation Intensity in Ice-Covered Lakes. December 2023.

[2] P. C. Matthews and S. I. Heaney. Solar heating and its influence on mixing in ice-covered lakes. *Freshwater Biology*, 18(1):135–149, August 1987.

[3] Benjamin J. Smith, Till J. W. Wagner, Hilary A. Dugan, Grace M. Wilkinson, Lucas K. Zoet, Nimish Pujara, and Jennifer A. Franck. How Ice Composition Controls Radiatively Driven Convection Under Lake Ice. *Geophysical Research Letters*, 52(19):e2025GL117454, October 2025.